

Availability Beyond Execution: Response-Oriented Throttling in Energy-Driven IoT

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Abstract. The Energy-Driven Computing (EDC) paradigm designs computation as a function of available energy, making it increasingly relevant to IoT deployments, such as sensor networks and robotic swarms, where devices must cooperate reliably under highly variable power conditions. Classical EDC, however, focuses solely on individual node energy survival, overlooking a second dimension of availability: when energy is depleted, a device loses both the ability to execute and to respond to its peers. This paper introduces software *throttling* as a pattern for addressing peer responsiveness in addition to energy survival. We distinguish the execution rate (ER) from the response rate (RR), where RR counts requests that receive an explicit denial, treating such rejections as valid responses. Across 729 scenarios evaluated in NS-3, we show that throttling exposes a controllable trade-off between ER and RR that an uncontrolled device cannot access. What a device monitors determines where in that trade-off it operates: monitoring the energy reserve favors responsiveness through conservation, while additionally monitoring harvesting capacity enables higher execution rates without sacrificing responsiveness. No strategy dominates both metrics simultaneously, and the gap between RR and ER widens as energy supply worsens. Each throttling design choice sustains measurable responsiveness to peers even when execution is no longer viable, a property that uncontrolled energy management cannot guarantee.

Keywords: throttling · energy-driven computing · service availability · Internet of Things · energy harvesting · responsiveness

1 Introduction

Energy-Driven Computing (EDC) paradigm designs computation as a function of available energy, rather than only the application’s requirements [17]. It is increasingly relevant to modern and future IoT applications, where large numbers of autonomous, distributed, and often energy-harvesting devices must operate reliably under highly variable power conditions. In IoT nodes, this constraint often stems from a reliance on harvesting energy from the environment, whose availability is finite, intermittent, and frequently unpredictable [11].

Highly interconnected IoT deployments, including robotic swarms, autonomous drone fleets, sensor networks, and collaborative robotic systems, depend on sustained cooperation among distributed nodes. Nodes that process every request without accounting for their current energy state may rapidly deplete their reserves and become unavailable, degrading both perceived quality of service and overall system performance.

To preserve operability under energy constraint, EDC systems restrict their activities according to the energy state, a behavior analogous to the software throttling pattern, defined as the deliberate limitation of operations to preserve a critical resource [9]. Energy-driven systems, however, treat the ability to tolerate a given energy context as the criterion of success. Under this view, energy exhaustion is understood merely as an interruption of operations.

There is, however, a dimension of availability that this framing does not address. When the reserve is exhausted, the device loses not only the ability to execute operations but also the ability to respond to the requester. From the perspective of whoever requests the service, a device that denies an operation but confirms the denial remains available, whereas a device that serves requests until it becomes unresponsive ceases to offer any guarantee. Distinguishing the ability to execute from the ability to respond reframes availability as a design problem, and it is precisely this distinction that throttling, by notifying the denial of operations, makes it possible to explore. We assume, as a design premise, that the explicit response has value for the requester, who can react to the denial by retrying, rescheduling, or redirecting the demand, rather than waiting out an expiration timeout.

This work evaluates throttling strategies derived from software patterns, taking service availability, rather than energy survival, as the central criterion of success. The strategies are compared through an experiment with four simulated devices and 729 scenarios run in NS-3, which separates the Execution Rate (ER), the operations actually executed, from the Response Rate (RR), the requests answered including notified denials.

This paper offers two contributions. The first is the framing of throttling oriented toward the responsiveness of an energy-driven system to its requester, and the second is the experimental evaluation of this framing, whose results indicate that throttling exposes a controllable trade-off between executing and responding, inaccessible to a system without control, and its reachable region is determined by the monitored observable.

Related work is presented in Section 2, followed by the foundation of throttling in the EDC context (Section 3), the experimental study (Section 4), the discussion of results (Section 5), and the conclusion (Section 6).

2 Related Work

Energy-driven IoT systems adapt their behavior according to the observed energy availability [17]. In this context, duty cycles are adopted to describe the time windows during which these systems remain active [11, 20]. Managing these

cycles is the central mechanism through which a system adjusts its behavior in the face of energy constraints. Merrett and Al-Hashimi [17] and Sliper *et al.* [19] observe that even devices able to tolerate energy intermittence require limiting mechanisms when the goal is to maintain some operational level, a situation in which the available energy is below the ideal yet still sufficient to sustain reduced duty cycles directed at critical activities, according to specific criteria.

A first class of approaches defines the behavior of duty cycles based exclusively on the instantaneous harvested power [17]. In these works, the harvested power observed at the moment of the decision serves as the determining factor for setting the system’s operating cycles [2–6, 13]. Although this strategy is simple and responsive, it is inherently reactive, since the system adjusts its behavior according to the instantaneous harvested power, without considering the accumulated state of the energy reserve or future demand.

To overcome this limitation, a second class of works incorporates the energy balance, understood as the relationship between harvested and consumed energy over a period, as a decision variable. Along this line, the capacity to perform operations is determined by the current drain on the energy reserve set against the harvested-energy readings [10, 12, 14, 16, 20, 21]. A second strand employs predictive and adaptive mechanisms. Ge and Nan [7] use reinforcement learning, and Shen *et al.* [18] machine learning, to adjust the duty cycle based on the estimated energy reserve; Lee and Park [15] and Arnaiz *et al.* [1] adopt adaptive control oriented toward an energy target. A third strand formulates energy management as an optimization problem, deriving decision policies over the system state [8, 22].

Across all the cited approaches, systems are designed to restrict their activities according to the energy state. This behavior is analogous to the throttling pattern described in the software systems literature [9]. In the related work, however, the resource to be preserved is always energy, and the criterion of success reduces to sustaining the system in terms of available energy reserve or to tolerating interruptions. None of the reviewed works addresses the system’s ability to respond to external requests as the target metric, leaving open the question of how throttling could be oriented by service availability rather than by the system’s energy survival alone. This work proposes that framing and empirically evaluates strategies derived from the identified design dimensions, taking service availability as the central criterion of success.

3 Throttling in the Energy-Driven Computing Context

The throttling pattern operates over three constitutive elements, namely (i) a finite resource whose exhaustion must be avoided, (ii) an observable threshold, and (iii) an execution demand to be regulated [9]. In energy-driven IoT systems, these three elements are present. The stored energy reserve is the finite resource, thresholds are defined by the energy state of the storage component, and the operation requests received by the access interface constitute the demand to be regulated [17, 19]. The difference between the two perspectives lies not in

the internal mechanisms but in what each makes visible to the designer. EDC designs these mechanisms with a focus on device survival, where the criterion of success is the energy balance [11, 20] or the ability to operate intermittently, tolerating energy-related constraints.

Framing the problem through throttling shifts the point of view toward the interface with the requester, where the criterion of success also comes to include whether the device was able to respond, whether by executing the operation or by explicitly notifying its refusal. A further distinction introduced by throttling is the use of explicit rejection instead of the absence of a response. This provides timely feedback that lets requesters adjust their behavior in collaborative ecosystems.

Given this new perspective, several design decisions must be considered according to the requirements of each application. The first concerns the *motivator*. The mechanism may be triggered to preserve the energy reserve within the observed operating limits, to actively restore that reserve when it falls below an acceptable level, seeking a positive energy balance until recovery, or to do both at once. The second decision involves the *threshold*, which may be fixed, operating with a constant and independent value, or adaptive, adjusting to continuous monitoring of the energy context. The third decision concerns the *observables*. The mechanism may be guided by the current energy reserve, by the harvesting capacity available in the environment, or by both in combination. The fourth decision defines the *means of action*. Restrictions may be applied based on the attributes of the requested operation (criticality, energy cost, periodicity), on the privileges of the requester, or through selective degradation of device components.

Thus, the strategies evaluated in the following section were derived from these four design decisions, combining them in distinct and controllable ways. This systematic derivation makes it possible to interpret performance differences not as accidental variations but as direct consequences of specific design choices, and to identify which combinations of decisions best preserve the system’s operation.

4 Experimental Study

To carry out the experiments, four IoT devices were simulated, each implementing a different throttling strategy, with three contrasting strategies (DISP2, DISP3, and DISP4) and one reference device (DISP1) that operates without any control mechanism. We evaluate the contribution of throttling strategies to service availability in energy-driven IoT systems. All four strategies were submitted to the same set of 729 experimental scenarios, run in NS-3, a discrete-event simulator.¹ This section describes the availability metrics adopted (Section 4.1), the configuration of the simulated devices (Section 4.2), the strategies evaluated (Section 4.3), the experimental setup (Section 4.4), and the results obtained.

¹ The complete results of the 729 scenarios and the analysis scripts are available in an anonymized repository at <https://anonymous.4open.science/r/5138> (the de-anonymized version will appear in the camera-ready).

4.1 Availability Metrics

In this study, availability is analyzed under two complementary aspects, (i) the ability to actually execute the requested operations and (ii) the ability to respond to the requester, whether by executing the operation or by a denial notification, without entering a state of total unavailability. This distinction has a practical implication, since a system that denies operations but retains the ability to communicate the denial is functionally more available than one that, after serving some requests, becomes unresponsive through energy exhaustion.

On the basis of this distinction, two metrics were adopted. The **Execution Rate (ER)** is the percentage of requests actually executed relative to the total number of requests received in a given scenario. The **Response Rate (RR)** is the percentage of requests to which the system responded relative to the total received, including denial responses, representing the availability perceived by the requester.

It is important to clarify that, when the system reaches the state of energy exhaustion (Depleted), it becomes unable to issue any response. Such requests are therefore counted as unavailable and penalize both the ER and the RR. Strategies that strongly restrict execution (low ER) but preserve the ability to respond (high RR) reveal a logic of energy self-conservation, whereas strategies with ER and RR close together operate with little differentiation between serving and denying.

4.2 Configuration of the Simulated Devices

Each simulated device represents an energy-driven IoT system with the ability to harvest energy from the environment. All devices share the same internal architecture, composed of four components.

The **Power Supply** component is the device's energy-harvesting component. It harvests energy from the environment and transfers it continuously to the Storage, operating independently of the state of the rest of the device. The **Storage** component retains the received energy up to its maximum capacity. Excess energy is wasted. When the Storage reaches zero, the system enters the Depleted state. The **Executor** component represents the performer of the operations authorized by throttling. In the active state, this Executor consumes energy from the Storage at a parameterized rate. In the waiting state, consumption is reduced but not zero, reflecting the minimum maintenance cost of a device that remains operational. The **Access Interface** is the entry point for external requests and the exit point for responses. Throttling is implemented adjacent to this interface, intercepting requests before they reach the Executor. The transmission cost is incurred by the Access Interface both when sending execution responses and when sending denial notifications.

The system's behavior is modeled by a state machine with three states, **Idle** (waiting, with minimum consumption), **Active** (executing operations, with high consumption), and **Depleted** (empty Storage). Transitions between states are

conditioned by the arrival of requests, the completion of operations, and the availability of energy in the Storage.

The energy dynamics follow this principle. The Power Supply transfers energy to the Storage continuously. The Storage provides energy to keep the system in the waiting state or to execute operations authorized by throttling. Based on the energy state of the Storage, a received request is executed, denied with a response to the requester, or, if the system is in the Depleted state, left without any response.

4.3 Throttling Strategies Evaluated

The four strategies evaluated were derived from the design dimensions presented in Section 3, combining them in distinct and controllable ways. All other parameters (architecture, storage capacity, and request arrival pattern) are kept equivalent across devices, so that the observed performance differences can be attributed exclusively to the throttling design choices.

DISP1 is the reference device, without a throttling mechanism. It attempts to execute every request received, regardless of its energy state, serving as a comparative baseline and representing the behavior of a system without availability control.

DISP2 implements throttling with a **fixed threshold and alternating action**. Its motivator is the reduction of average energy consumption through uniform load control, without reference to the energy state. The action consists of serving one out of every two requests received, denying the next in an alternating fashion, without observing the energy state, the criticality of the operations, or the privileges of the requester. The throttling monitors no observable and operates statically, modeling a flow control insensitive to the energy context.

DISP3 and **DISP4** adopt an adaptive threshold based on four **operating modes**. The modes represent ranges of the percentage of energy remaining in the Storage and determine the degree of restriction that throttling applies to each request received. As the energy reserve decreases, the system transitions to progressively more restrictive modes, from the **Abundant** mode, characterized by full energy availability, to the **Hibernating** mode, of minimal reserve, bordering the Depleted state. The thresholds of each mode and the expected action of DISP3 and DISP4 are described in Table 1.

Table 1. Expected action for devices DISP3 and DISP4

Mode	Energy Range (S)	Allowed Operations
Abundant	$S \geq 70\%$	100%
Attention	$70\% > S \geq 35\%$	80%
Alert	$35\% > S \geq 10\%$	50%
Hibernating	$10\% > S \geq 0\%$ *	0%

* On reaching 0%, the system enters the Depleted state.

Table 2. Throttling strategies evaluated

Device Throttling		Action Characteristics		
		Threshold	Mechanism	Observables
DISP1	No	–	–	–
DISP2	Yes	Fixed	Operation rate	–
DISP3	Yes	Adaptive	Operating mode	Energy reserve
DISP4	Yes	Adaptive	Operating mode	Harvesting and energy reserve

DISP3 implements throttling with an **adaptive threshold oriented toward conservation of the energy reserve**. Its motivator is the preservation of the reserve within operating limits. The observable is the current energy reserve, evaluated at each duty cycle estimated at 1 second. The means of action are proportional to the observed mode, as shown in Table 1. In the Hibernating mode, no operation is executed, but the system preserves the ability to issue denial notifications to the requester.

Whereas DISP3 monitors solely the energy reserve, DISP4 also monitors the instantaneous harvested power. This difference changes the behavior between the two in the Hibernating mode. Without access to harvesting information, DISP3 recovers the reserve by fully blocking the execution of operations. DISP4, by observing the instantaneous harvested power, allows operations per duty cycle to the extent that the harvested energy sustains them, retaining a margin for gradual recovery of the reserve. Thus, DISP4 keeps the system partially responsive to the requester during energy recovery.

4.4 Experimental Setup and Control

The experiment followed four explicit control criteria, (I) all systems were specified according to the design dimensions established in Section 3; (II) all have equivalent energy storage capacity; (III) all were submitted simultaneously to the same number of requests in each scenario; (IV) the energy resource was provided equitably to all devices according to each scenario. The independent variables comprise six dimensions that reflect characteristics typical of IoT systems operating from their energy-harvesting capacity, with the ranges of variation summarized in Table 3.

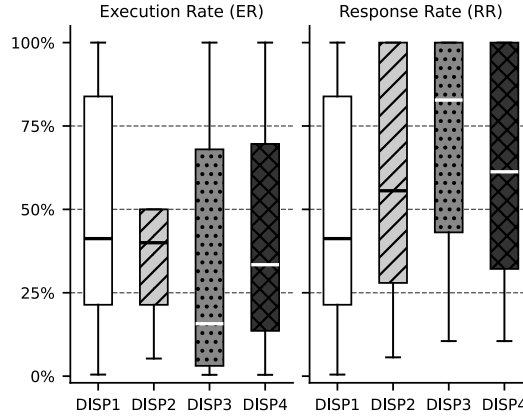
Each request is recorded with one of the following outcomes, **served** (executed successfully), **denied** (refused by throttling, with notification to the requester), or **unavailable** (system in the Depleted state, unable to respond).

4.5 Overall Results

Figure 1 makes the central effect of throttling evident. In all devices with a control mechanism, the RR box sits systematically above the ER box, indicating that the system preserves the ability to respond even when it restricts execution.

Table 3. Independent variables of the experiment

Variable	Minimum	Intermediate	Maximum
<i>Energy context</i>			
harvest value	5 J/cycle	10 J/cycle	20 J/cycle
buffer size	300 J	1200 J	4800 J
<i>Operational costs</i>			
idle cost	0.05 J	0.6 J	1.2 J
active cost	0.15 J	0.9 J	1.5 J
transmission cost	0.2 J	1.1 J	1.4 J
<i>Interval</i>			
request interval	0.05 s	0.10 s	0.20 s

**Fig. 1.** Distribution of ER and RR across the 729 scenarios for each strategy

DISP1, without any control, exhibits the pattern opposite to the others. Its ER and RR distributions are identical (mean of 49.57%), because the device serves while there is reserve and falls silent upon exhausting it, making executing and responding inseparable. DISP2, with fixed throttling, has its ER capped at 50% in every scenario (a ceiling visible in the upper whisker of the box) and an RR with a bimodal distribution (344 scenarios below 50%, 297 above 75%). It operates well or poorly depending on the energy context, without stable intermediate states.

Among the adaptive-threshold systems, DISP3 shows the greatest divergence between ER and RR (430 scenarios with ER below 25% and 406 with RR above 75%), an effect of the full blocking of execution in the Hibernating mode, which still preserves the notification to the requester. DISP4 is the most balanced, with the ER and RR boxes closer together and more scenarios with ER above

50% (266) than the other controlled strategies, because active recovery allows operations proportional to harvesting instead of blocking entirely.

Read together, ER and RR place each system as a point in the trade-off plane between executing and responding, rather than as positions in a single ordering. DISP1, without control, remains on the diagonal where ER and RR coincide (median of 41.2% for both), because for it executing and responding are the same capability. The systems with throttling shift this point above that diagonal, converting the restricted executions into the ability to respond, DISP2 to (40.0%; 55.6%), DISP4 to (33.4%; 61.3%), and DISP3 to (15.7%; 82.8%), in medians of ER and RR. No system dominates the others on both coordinates, since a higher RR comes at the cost of a lower ER, so the comparison describes a trade-off, not a winner. The RR, therefore, is not interpreted on its own; it is a coordinate whose value depends on how much execution one accepts to give up in order to preserve the response. The difference introduced by throttling thus manifests in the ability to respond to the requester, and not in the ability to execute operations, reflecting the trade-off described above.

The simulation is deterministic. Requests arrive periodically, spaced by the request interval over a fixed horizon of 600 seconds, so that each scenario produces a unique and reproducible result, which is confirmed by the identity of the results across the three sessions. The dispersion observed in the boxes does not, therefore, reflect sampling variability, but the sensitivity of each strategy to the factorial grid of 729 scenarios. Even so, the difference between the strategies is systematic. The Friedman test rejects the hypothesis of equivalence both for the ER ($\chi^2(3) = 880.3$, $p < 0.001$, Kendall's $W = 0.40$) and for the RR ($\chi^2(3) = 1407.8$, $p < 0.001$, $W = 0.64$), with stronger agreement in the ordering by ability to respond. In the pairwise comparison (Wilcoxon with Holm correction), DISP3 and DISP4 surpass DISP1 in RR with maximum effect size (rank-biserial $r = 1.0$, $p < 0.001$) in all scenarios, whereas, in ER, it is DISP1 that maintains an equally systematic advantage.

4.6 Correlations between Independent Variables and Availability

The relationship between the six independent variables and the availability metrics was quantified by the Spearman correlation coefficient (r_s), presented in Figure 2 for each variable and system. Since the experiment exhaustively traverses the 3^6 factorial grid, the correlation does not estimate a population parameter but measures the monotonic sensitivity of availability to each factor within the space of evaluated scenarios. Unless otherwise indicated, the reported coefficients are significant at $p < 0.001$, with 95% confidence intervals (bootstrap, 5000 resamples) narrower than 0.08; the only exceptions are the correlations of active cost with the RR of DISP3 and DISP4, statistically indistinguishable from zero.

The energy-context variables correlate positively with availability. The harvest value has the strongest effect on the ER of DISP4 ($r_s = 0.59$), whose active recovery depends on harvested energy to resume operations, and on the ER of DISP1 ($r_s = 0.49$), which consumes that energy until exhaustion. The buffer size correlates more weakly, with a maximum effect on the ER of DISP3 ($r_s = 0.42$),

	Execution Rate (ER)				Response Rate (RR)			
harvest value	+0.49	+0.49	+0.34	+0.59	+0.49	+0.44	+0.49	+0.35
buffer size	+0.28	+0.15	+0.42	+0.24	+0.28	+0.28	+0.13	+0.22
idle cost	-0.18	-0.20	-0.23	-0.22	-0.18	-0.20	-0.29	-0.20
active cost	-0.20	-0.15	-0.14	-0.23	-0.20	-0.14	-0.02	-0.06
transmission cost	-0.37	-0.39	-0.38	-0.36	-0.37	-0.37	-0.37	-0.41
request interval	+0.60	+0.60	+0.56	+0.48	+0.60	+0.62	+0.57	+0.68
	DISP1	DISP2	DISP3	DISP4	DISP1	DISP2	DISP3	DISP4

Fig. 2. Spearman coefficients (r_s) between independent variables and availability. Gray intensity proportional to $|r_s|$; sign indicated in the cells.

because larger reserves prolong the stay in less restrictive modes. In DISP1 and DISP2, which do not modulate behavior by the reserve, storage capacity has little influence.

The operational costs correlate negatively. The transmission cost has the greatest impact (r_s between -0.36 and -0.41 across all systems), because even denial notifications incur it, which reveals a limit of throttling, namely that the explicit response competes for the resource it aims to preserve. The idle cost has a moderate and uniform effect (-0.18 to -0.29), being paid continuously and not mitigable by throttling. The active cost is the weakest and, for the RR of DISP3 and DISP4, practically null (-0.02 and -0.06), because the adaptive threshold reduces active operations in the restrictive modes and decouples the response from that cost.

The request interval has the highest positive correlation (ER between 0.49 and 0.60 , RR between 0.57 and 0.68), with a maximum effect on the RR of DISP4 ($r_s = 0.68$), sensitive to the spacing between requests because short intervals pressure the reserve before recovery completes. These coefficients do not prescribe ideal values; they describe how availability responds to each condition. The design implications are taken up in Section 5.

4.7 Evaluation by Independent Variable

To assess the effect of each independent variable, one is varied at a time while the others remain at their intermediate values. The operational costs (idle cost, active cost, and transmission cost) determine the pace of energy consumption and, together, define three scenarios, as described in Table 3.

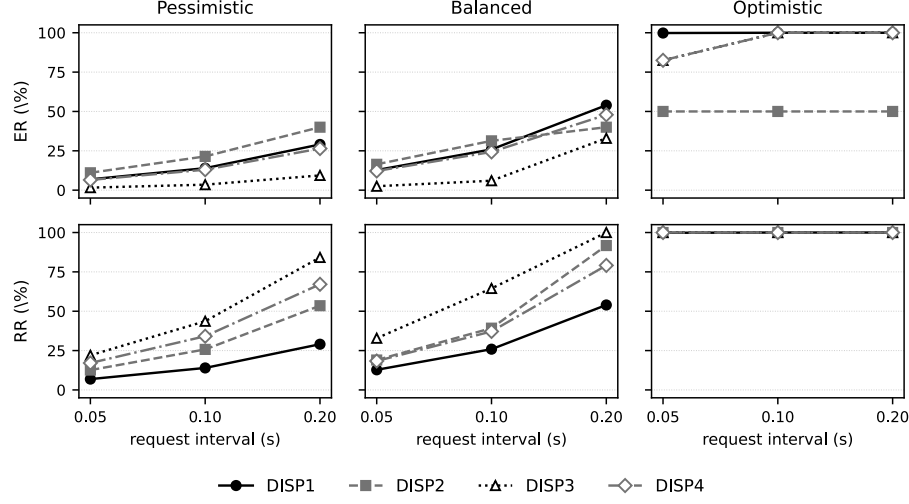


Fig. 3. Availability (ER and RR) of the four systems according to the request interval, in the three operational-cost scenarios.

Pessimistic scenario operational costs at their maximum values, representing systems in which operations and responses consume significant energy, characteristic of devices with long-range transmission or intensive processing.

Balanced scenario intermediate operational costs, representing typical operating conditions.

Optimistic scenario minimum operational costs, representing systems with low cost per operation.

Variation of the Request Interval Smaller request intervals raise the number of operations per duty cycle and accelerate consumption of the reserve. Figure 3 presents the ER and RR of the four systems across the three cost scenarios.

The effect of demand depends on the adversity of the scenario. In the optimistic scenario, energy absorbs the demand and the systems maintain near-full availability, except for the 50% ceiling of DISP2, and only at the smallest interval does the ER of DISP3 and DISP4 recede, with the RR preserved at 100%. In the pessimistic scenario, reducing the interval collapses the ER of all systems, but the adaptive-threshold ones sustain the highest RR while DISP1 and DISP2 lose ER and RR in the same proportion. The balanced scenario exposes the contrast. At a long interval, DISP3 reaches an RR of 100% with an ER of 33%, and DISP4 combines an ER of 48% with an RR of 79%, whereas DISP1 equals ER and RR at 54%.

Across all intervals, the RR of DISP3 and DISP4 remains above that of DISP1, and the distance between them grows as the interval decreases and execution becomes infeasible.

Table 4. Availability (ER and RR, in %) by energy context, in the pessimistic and balanced scenarios

System	harvest value (J/c)						buffer size (J)					
	5		10		20		300		1200		4800	
	ER	RR	ER	RR	ER	RR	ER	RR	ER	RR	ER	RR
<i>Pessimistic scenario</i>												
DISP1	4.0	4.0	14.0	14.0	33.9	33.9	11.0	11.0	14.0	14.0	26.0	26.0
DISP2	11.8	15.5	21.4	25.8	40.0	46.7	20.4	21.4	21.4	25.8	25.8	43.1
DISP3	3.1	24.1	3.5	43.6	4.8	82.0	0.9	40.8	3.5	43.6	13.9	54.9
DISP4	3.1	24.1	13.0	34.1	32.8	44.8	10.8	31.0	13.0	34.1	21.9	46.8
<i>Balanced scenario</i>												
DISP1	15.6	15.6	25.9	25.9	55.8	55.8	21.4	21.4	25.9	25.9	43.6	43.6
DISP2	22.0	28.0	31.3	39.2	40.0	81.8	30.3	32.2	31.3	39.2	35.3	67.2
DISP3	4.9	35.9	6.0	64.5	24.0	100.0	1.5	61.0	6.0	64.5	23.8	78.4
DISP4	4.9	35.9	24.3	37.2	60.8	68.0	21.1	31.7	24.3	37.2	36.9	59.3

In the optimistic scenario (minimum operational costs), availability saturates at 100%, except for the 50% ER ceiling of DISP2.

Variation of the Energy Context The energy context comprises harvest value and buffer size, which affect the energy reserve in different ways and are therefore evaluated in isolation before the joint analysis.

The harvest value (5 to 20 J) and the buffer size (300 to 4800 J) raise the availability of the two adaptive systems (Table 4), but through distinct observables, which differentiates them. With minimal harvesting, DISP3 and DISP4 coincide (ER of 3.1% and RR of 24.1% in the pessimistic scenario); as harvesting grows, they diverge, because only DISP4 observes it and converts it into execution (ER of 60.8% in the balanced scenario, above DISP1), whereas DISP3 converts it into response (RR of 100%). The buffer size, by contrast, acts on the reserve that both observe and raises the availability of the two without distinguishing them, with a proportionally larger ER gain in DISP3 (about sixteen times in the balanced scenario). The systems without an adaptive threshold merely track the energy context, DISP1 until exhaustion and DISP2 within its 50% ER ceiling, and the effect of both parameters vanishes in the optimistic scenario. The energy context, therefore, translates into availability only where throttling observes the reserve, and the distinguishing feature of DISP4 comes from also observing harvesting.

5 Discussion

In every system with control, the RR sat above the ER, which repositions each system outside the scenario where executing and responding fully coincide. This separation between responding and executing, inaccessible to the system without control, is the core of the contribution and organizes the discussion of the

findings. Yet a greater ability to respond is obtained at the cost of less execution, so that throttling does not maximize a metric but makes the operating point between the two controllable.

The reference without throttling, DISP1, has no control mechanism, and its execution and response rates are inseparable. The device executes while there is reserve and falls silent once it is exhausted, so that losing the ability to execute entails losing the ability to respond as well. The systems with throttling dissociate the two capabilities, deliberately sacrificing execution to preserve the response. From the requester’s perspective, a device that denies an operation but confirms the denial remains available, whereas a device that serves until it becomes unresponsive ceases to offer any guarantee. It is this perceived availability, and not merely the throughput of operations, that response-oriented throttling protects.

This concern becomes more relevant as modern systems take on an increasingly distributed and collaborative character, as in drone swarms, connected vehicles, smart cities, and monitoring networks. In these scenarios, the ability to signal the momentary inability to serve a request, letting the requester adjust its demand behavior, is an essential requirement for sustainable collaboration.

We assume here that this response has value for the requester, who, on receiving an explicit denial, can react, whereas silence forces it to wait out the expiration. The experiment does not model the requester, so this value is taken as a premise, not as a measured result.

The adaptive-threshold systems, DISP3 and DISP4, adjust restriction according to the reserve state and occupy distinct points, both above the uncontrolled system. Their difference lies in DISP4’s additional observable. DISP3 monitors only the energy reserve and therefore conserves under scarcity, minimizing execution and converting available energy into responsiveness, reaching a high RR with a near-zero ER. This preserves responses but nearly eliminates execution, representing a design limit rather than a recommended configuration. DISP4 also monitors harvesting capacity and, in Hibernating mode, allows execution proportional to harvested energy instead of blocking it entirely, preserving a margin for reserve recovery and achieving the best balance between execution and responsiveness.

The effect of the adaptive threshold, against the reference without control, is not uniform but proportional to the adversity of the conditions. Under low costs or reduced demand, the available energy absorbs the load and all systems converge to near-full availability, a situation in which throttling adds little and the fixed threshold of DISP2 is even detrimental. It is under energy scarcity and a high frequency of requests, when execution becomes infeasible and the absence of control would lead to total unavailability, that the adaptive systems stand out by sustaining the response.

The operational costs delimit the reach of throttling. The adaptive threshold practically decouples the ability to respond from the active cost, by restricting active operations in the more restrictive modes, but it can do nothing against

the idle cost, paid continuously, which establishes an inescapable floor of consumption.

The transmission cost exposes an intrinsic limit of the pattern. It was the cost of greatest impact on both metrics (r_s between -0.36 and -0.41), and the reason is structural. The denial notification, the mechanism that defines response-oriented throttling and preserves the RR, incurs a cost similar to that of transmitting an execution response. In environments where transmitting is expensive, denying frequently may consume more energy than it saves, so that the explicit response competes for the resource it aims to preserve. The transmission cost should therefore be treated as a design variable, not as a fixed operating parameter, calibrating the frequency of denials according to the cost of issuing each notification.

5.1 Limitations

The strategies evaluated are not optimal configurations; the observed values describe how availability responds to each condition, not a target to pursue. The mode thresholds (70%, 35%, and 10% of the reserve) partition the reserve into qualitatively distinct regimes and were not calibrated for a specific energy profile, so adjusting them, such as shifting the Attention and Alert boundary, may produce different results, a sensitivity that is itself a design variable.

The comparison took DISP1 as the sole reference. Omitting a baseline from the EDC literature is deliberate, since the reviewed approaches target the maintenance of execution rather than the ability to respond, so confronting them directly on RR would compare metrics serving different purposes. Absent such a comparator, the three strategies still traverse the design space between context-insensitive control and adaptive control with active recovery, without exhausting the possible ones. The cost of the throttling logic itself, though typically lower than that of the controlled operations, is not zero and must be considered in real implementations.

Finally, the experiment was simulation-based, and the fidelity of the consumption and harvesting models relative to physical devices remains a limitation. Because all four strategies were evaluated under identical conditions, the comparisons are internally consistent and the relative findings do not depend on absolute model accuracy. Simulation also made the evaluation tractable: running 729 scenarios on physical devices would be unfeasible, so the broader coverage represents a deliberate trade-off against deployment realism.

6 Conclusion

This work investigated the throttling pattern as a mechanism to improve the availability of energy-driven IoT systems, from the perspective of the ability to respond to the requester. Through an experiment with four simulated devices and 729 scenarios, we showed that throttling produces an availability profile qualitatively distinct from that of energy-driven system patterns, which center on

maintaining execution. By distinguishing the ability to execute operations (ER) from the ability to respond to the requester (RR), the systems with throttling make the trade-off between the two controllable, shifting the operating point beyond the condition where executing and responding coincide and offering the ability to respond where the absence of control would lead to energy exhaustion and, with it, to the loss of the ability to respond itself.

The results show that this control capability is not uniform but determined by the throttling design choices, above all by the observable it monitors. A fixed threshold, insensitive to the energy state, controls execution predictably but without margin to adapt to the scenario. An adaptive threshold, by contrast, converts the energy state it observes into the ability to respond, and the reach of this conversion depends on what is observed, since observing the reserve allows conserving it, whereas also observing the harvesting capacity allows converting energy into execution without giving up the response. The benefit of these systems grows with the adversity of the conditions, being greatest precisely where execution becomes infeasible. The pattern has, however, an intrinsic limit, since the denial notification that characterizes it competes for the same transmission cost it aims to save.

The value of the explicit response for whoever requests the service was taken as a design premise, not measured. Modeling the requester's behavior in future work would make it possible to quantify this benefit and calibrate the frequency of denials against the cost of notifying.

Even so, the central result remains. By treating availability as the ability to respond to the requester, and not only to execute, throttling offers designers of energy-driven systems a means to preserve the relationship with the requester even under the conditions in which execution ceases to be viable.

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